Application of Information & Communication Technologies

Lecture-11

Recap of Lecture 10

- Output Devices
- How computers give results back to us

Overview of Lecture 11

- System vs Application Software
  - Differences
  - Functions of OS
  - Types of OS

What is Software?

- Software = instructions/programs for the computer.
- Two main categories:
  - System Software
  - Application Software

System Software

- Helps the computer itself run.
- Controls computer hardware.
- Works in the background.
- Needed before apps can run.
- Examples: Windows, Linux, MacOS.

Application Software

- Helps the user do tasks.
- Runs on top of system software.
- Examples: MS Word, Edge Browser, Map.

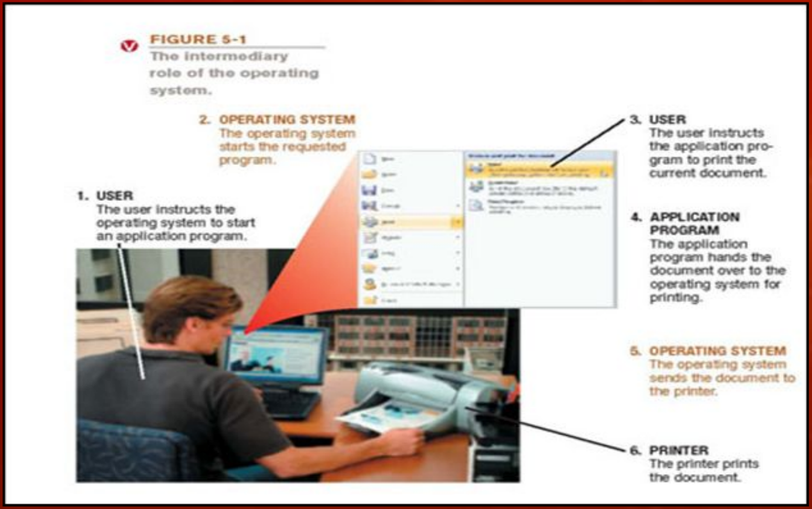
Key Differences

Feature	System Software ⚙️	Application Software 🛠️
Purpose	Run computer	Help user tasks
Works in	Background	Front-end (user runs it)
Dependency	Needed before apps	Needs system software
Examples	Windows, Linux	Word, Excel, Photoshop

- **Easy Analogy**
  - **System Software** = Engine → makes car run.
  - **Application Software** = Steering/Tools → used by driver.

Operating System (OS)

- A collection of programs that manage and coordinate the activities taking place within a computer system
- Acts as an intermediary between the user and the computer
- Main type of system software.
- Connects: User ↔ Applications ↔ Hardware.
- Example: Windows, MacOS, Linux, Android.



Functions of an Operating System

- Device Management (printers, drives).
- Memory Management (RAM usage).
- Process Management (CPU tasks).

- File Management → create, delete, organize.
- User Interface → GUI (desktop) or CLI (commands).
- Security → passwords, permissions.

- Multitasking → run many apps at once.
- Communication → networking, internet access.

Functions of an Operating System

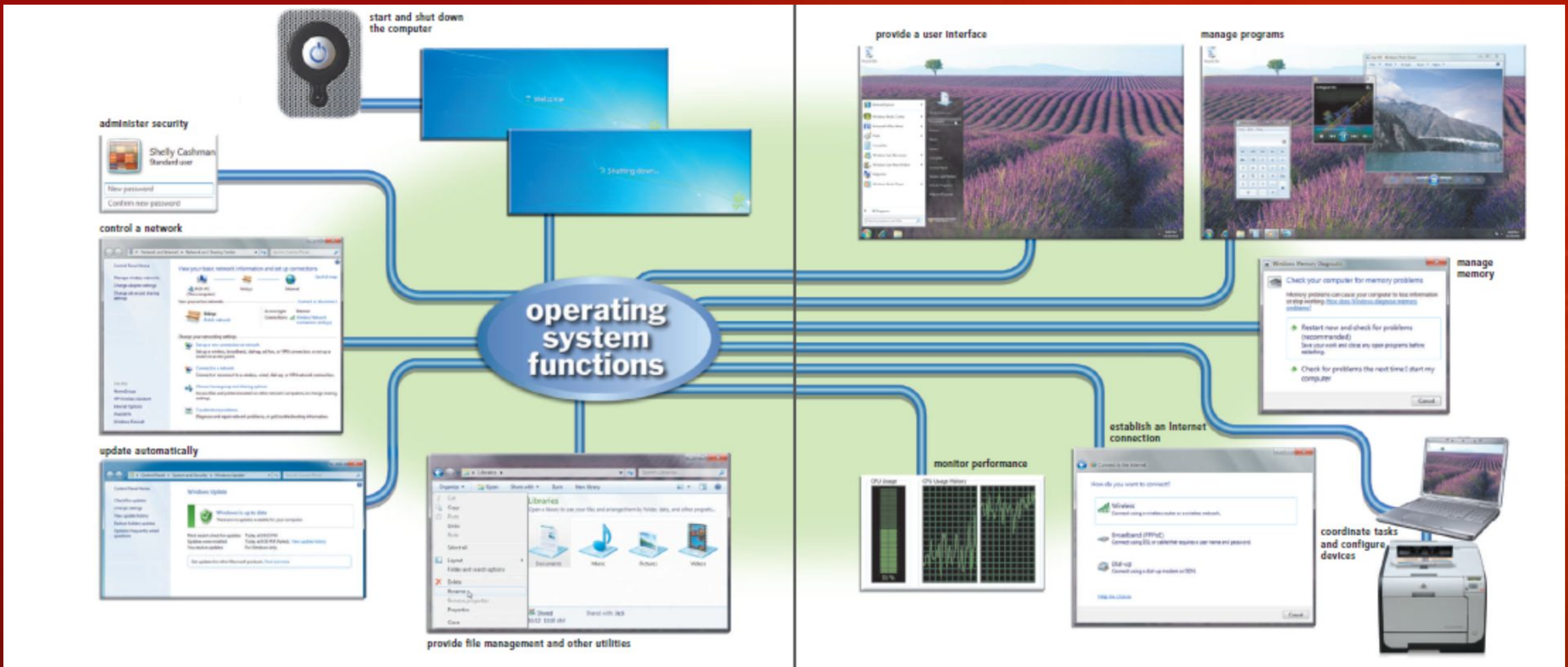


Figure 7-1 Most operating systems perform similar functions, which are illustrated with the latest version of Windows in this figure.

File Management

- File Management → create, delete, organize.

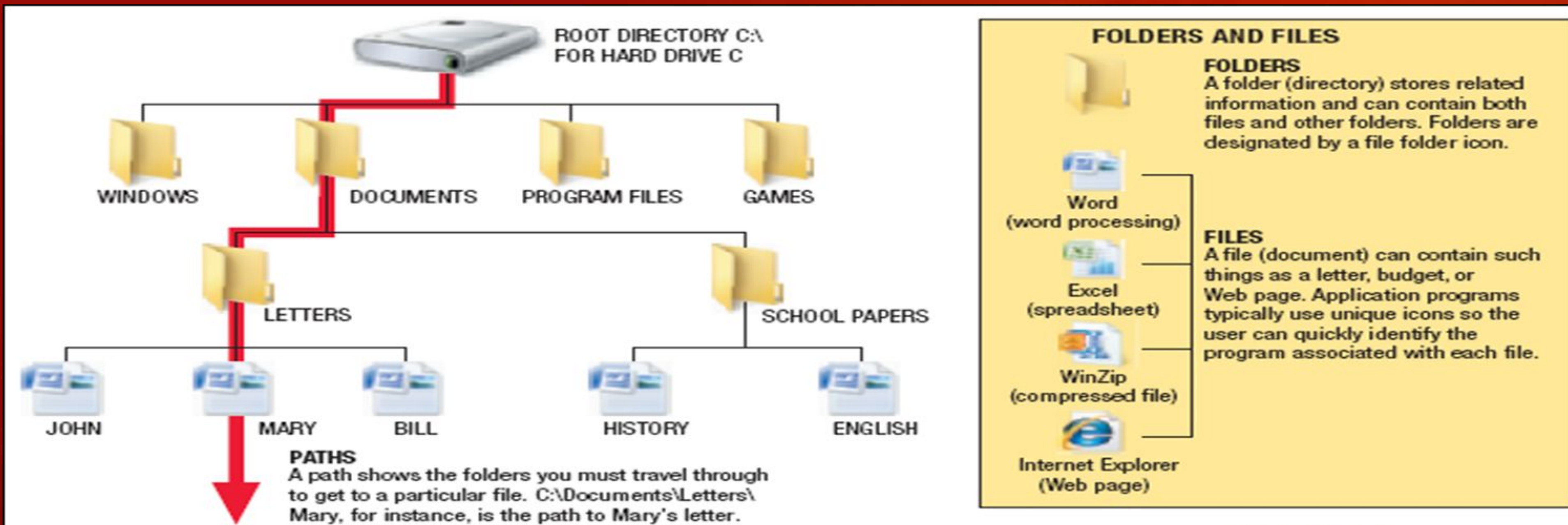


FIGURE 5-5
A sample hard drive organization.

Processing Techniques for Increased Efficiency

- Memory management: Optimizing the use of main memory (RAM)
  - Virtual memory: Memory-management technique that uses hard drive space as additional RAM

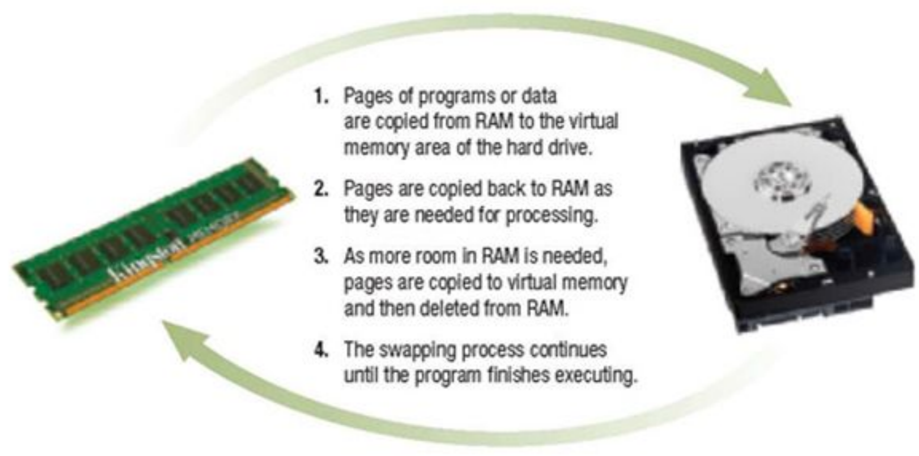
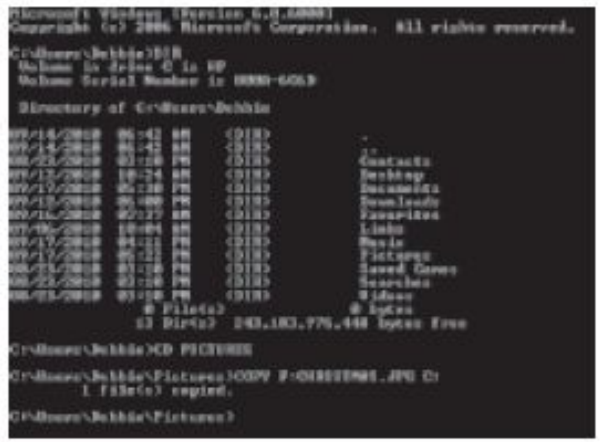


FIGURE 5-8
How virtual memory works.

Differences Among Operating Systems

- Command line vs. graphical user interface (GUI)
  - Most operating systems use GUI today

GRAPHICAL USER INTERFACE
Icons, buttons, menus, and other objects are selected with the mouse to issue commands to the computer.



COMMAND LINE INTERFACE
Commands are entered using the keyboard.

FIGURE 5-10
Command line vs. graphical user interfaces.

Differences Among Operating Systems

- Personal and server operating system
  - **Personal operating system:** designed to be installed on a single computer and support one user at one time
  - **Server operating system:** designed to be installed on a network server
    - Client computers still use a personal operating system
    - Server operating system controls access to network resources
    - *Many users share resources.*
  - Many operating systems come in both versions

Server Operating Systems

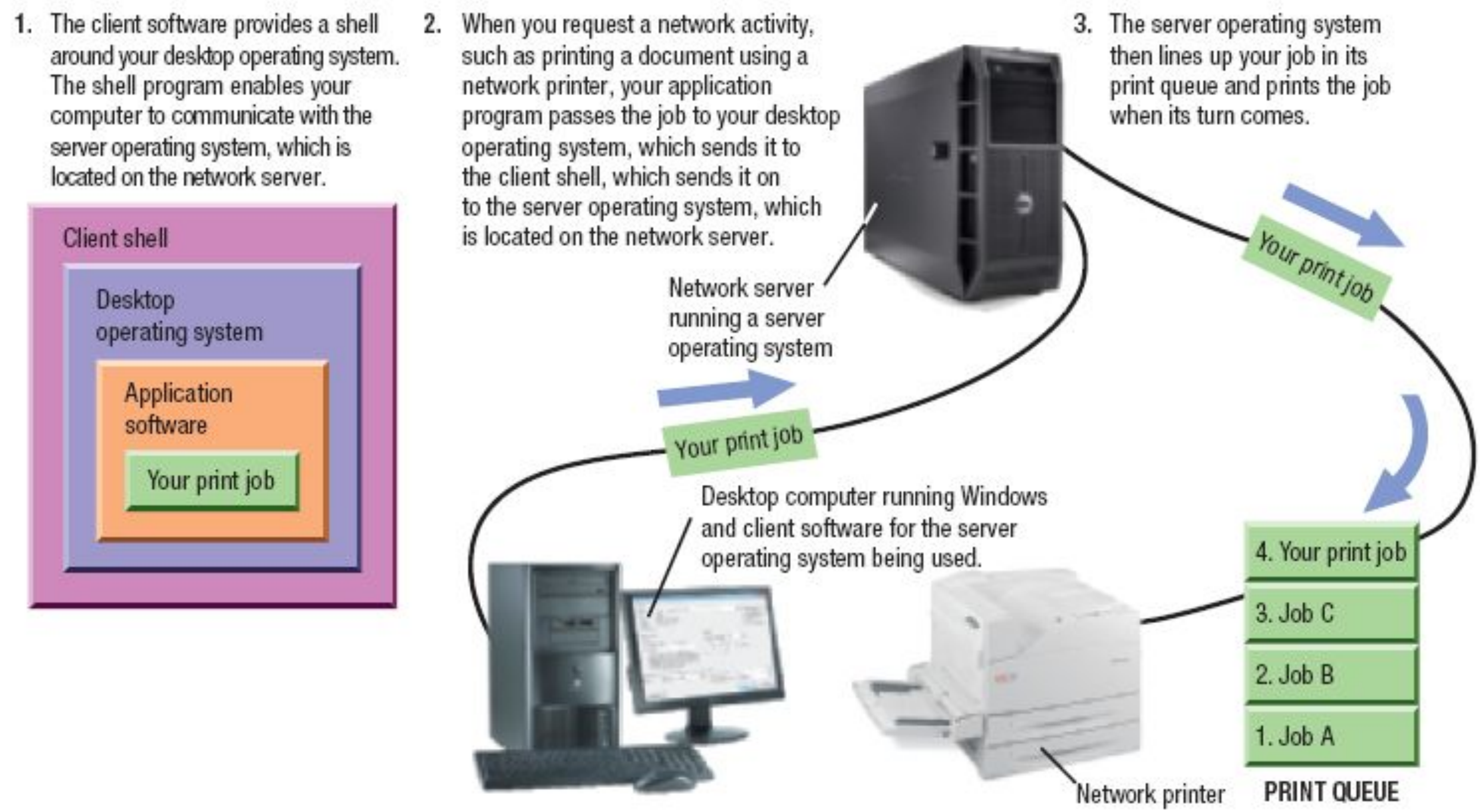


FIGURE 5-11
How operating systems are used in a network environment

OS's *for* Personal Computers and Servers

- **DOS: Disk Operating System**
  - **PC-DOS:** Created originally for IBM microcomputers
  - **MS-DOS:** used with IBM-compatible computers.
  - DOS traditionally used a command-line interface.
  - Can enter DOS commands in Windows.

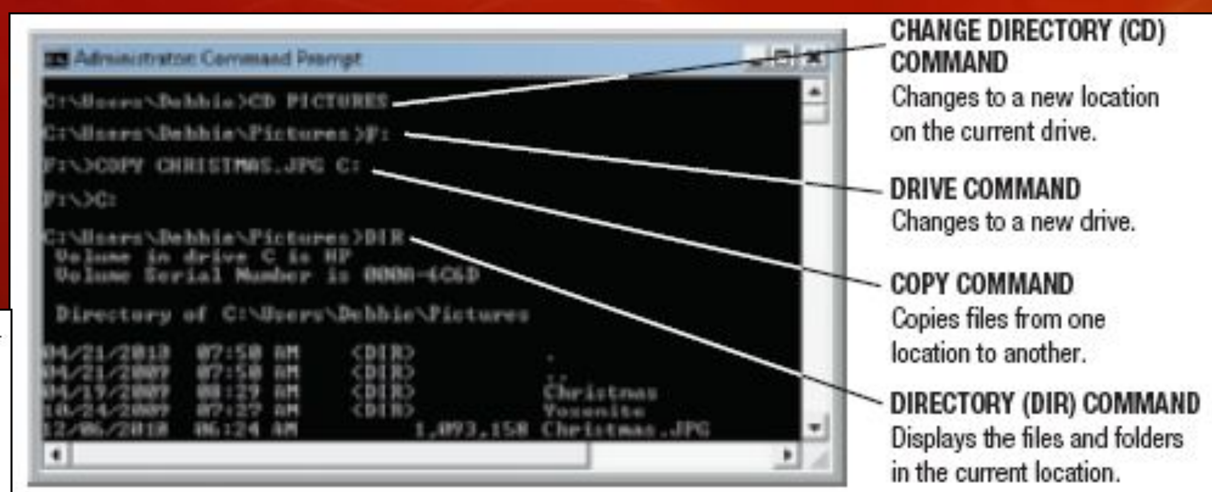


FIGURE 5-12
DOS. Even though DOS has become technologically obsolete, Windows users can still issue DOS commands via the Command Prompt.

Microsoft Windows OS

- Windows = Most widely used personal computer OS by Microsoft
- Graphical User Interface (GUI) made computers easy to use
- Versions evolved with better speed, security, multimedia, and networking

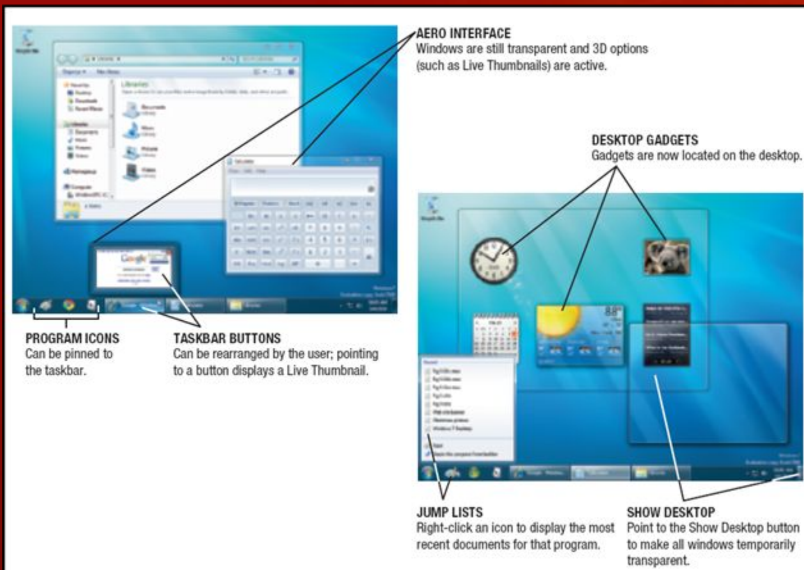


FIGURE 5-14
Windows 7.

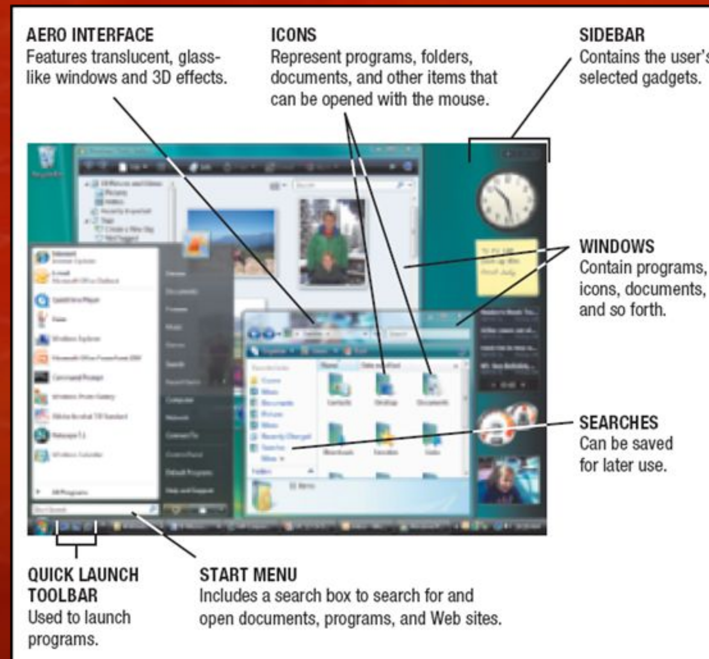


FIGURE 5-13
Windows Vista.

Evolution of Windows

Early Versions

- **Windows 1.0** → 3.x: Added GUI to DOS
- **Windows 95 / 98:** Improved graphics, Internet support
- **Windows NT / 2000:** 32-bit architecture, stable for servers & businesses
- **Windows Me** (Millennium Edition): Home networking & multimedia

Modern Versions

- **Windows XP:** Very popular, multimedia & networking improvements
- **Windows Vista:** New Aero design, security, search features
- **Windows 7:** Faster, stable, better networking & device ssupport
- **Windows 8 / 8.1:** Touch interface, tiles, cloud sync
- **Windows 10:** Universal apps, Cortana, Microsoft Edge browser

Windows Servers & Newer Versions

- **Windows Server** (2008 → 2019): Designed for enterprise & cloud computing
- **Windows 11 (latest):** Modern design, better performance, optimized for hybrid work
- **Supports** cloud integration, security, Android apps (via Amazon store)

Comparison Table

Version	Year	Key Feature
Windows 95	1995	Plug & Play, GUI improvements
Windows XP	2001	Stable, multimedia
Windows 7	2009	Fast, user-friendly
Windows 8	2012	Touch & tiles interface
Windows 10	2015	Universal apps, Edge browser
Windows 11	2021	Modern UI, hybrid work tools

UNIX

- UNIX: Operating system developed in the late 1960s for midrange servers
  - Multiuser, multitasking operating system
  - Stable and secure → widely used in servers and research
  - Basis for many modern OS (Linux, Mac OS, Android)
  - Popular versions: Solaris, AIX, HP-UX

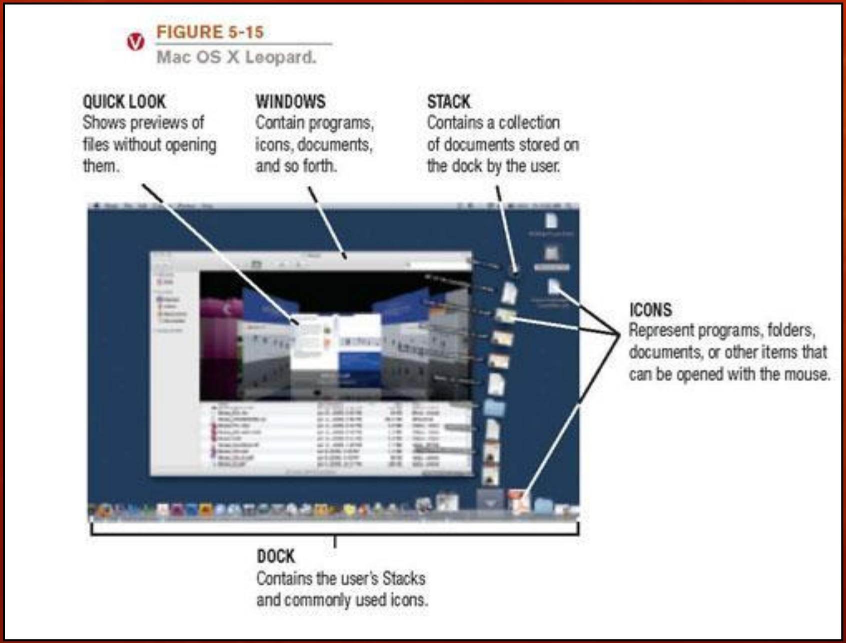
Linux

- Linux: Version (flavor) of UNIX available without charge over the Internet
  - Open-source version of UNIX
  - Free to use and modify → supported by a large community
  - Runs on PCs, servers, smartphones, embedded systems
  - Distributions: Ubuntu, Fedora, Red Hat, Debian, Kali Linux
  - Known for: security, customization, performance
  - Powers most of the internet servers & supercomputers

FIGURE 5-16
Linux. This version of Linux includes a 3D graphical interface to increase the user's workspace.

Mac OS

- Mac OS: Proprietary operating system for computers made by Apple Corporation
  - Based on the UNIX operating system; originally set the standard for graphical user interfaces
  - Mac OS X Snow Leopard: Most recent personal version
  - Includes:
    - Safari Web browser
    - New features like Time Machine, Stacks, Quick Look, Boot Camp, etc.
    - More responsive than previous versions



Comparison of OS Families

OS Family	Key Strengths	Typical Users
Windows	Easy to use, wide compatibility	General users, businesses
UNIX	Stable, secure, scalable	Servers, research, enterprise
Linux	Open-source, customizable	Developers, servers, cybersecurity
Mac OS	Creative tools, user-friendly	Designers, multimedia professionals

Summary

- System software makes the computer run.
- Application software makes it useful for us.

Suggested Reading

- Ch-05, The System Unit: Processing and Memory , “Understanding Computers: Today and Tomorrow, Comprehensive”, 15th Edition by Deborah Morley & Charles S. Parker

- Ch-07, Discovering Computers Fundamentals- Your Interactive Guide -- Gary B Shelly; Misty E Vermaat; Jeffrey J Q